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Goal-directed action planning modulates surround suppression in the visual system

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38 **Author contribution statement**

39 Y.Z. and J.G: Conceptualization, Formal analysis, Methodology, Software, Visualization,
40 Writing - original draft, Writing - review & editing. Z.D., T.W., W.W., Y.W., C.W., X.L., Y.W:
41 Data curation. J.L and B.H: Formal analysis, Methodology; J.R.F and J.G:
42 Conceptualization, Writing – review & editing. J.C: Conceptualization, Funding acquisition,
43 Project administration, Methodology, Resources, Supervision, Writing - Original draft,
44 Writing - review & editing.

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54 **Abstract**

55 Action planning has been traditionally viewed through the lens of motor command
56 specification. However, converging evidence indicates planning also modulates early
57 cortical sensory systems. While previous studies mainly focused on the modulation to a
58 single target object, it remains unclear how action planning modulates the spatial interaction
59 between target and flankers in a cluttered scene. Here we directly tested this using EEG
60 with frequency tagging to measure visual surround suppression — a key cortical
61 mechanism for object segregation — while participants (22 females, 14 males) planned to
62 grasp a target among flankers. We hypothesized that surround suppression, essential for
63 distinguishing the target from distractors, would be enhanced when planning to grasp the
64 target compared to planning a non-target-directed action where segregation is less critical.
65 Further, based on evidence suggesting that surround suppression when *executing* a
66 grasping movement is greater for dominant than nondominant hand, we predict that this
67 specificity should likewise exist during *planning*. Confirming our predictions, EEG results
68 demonstrated that planning a grasp significantly amplified surround suppression relative to
69 planning a simple key release. This enhancement was additively modulated by target
70 relevance and hand dominance, being strongest when planning a dominant-hand grasp.
71 Crucially, a control task (planning a perceptual judgment) elicited no such suppression,
72 supporting the action-specificity of this sensory tuning. These results provide neural
73 evidence at the level of visual mechanisms, that action preparation actively and selectively
74 refines early sensory processing for efficient goal-directed action.

75 **Significance**

76 The neural mechanisms of the top-down modulation of the motor system on the sensory
77 system during action planning remain unclear. Previous studies mainly focused on
78 modulation during action on a single object. Here, we demonstrate that grasp planning

modulates surround suppression to enhance edge detection and facilitate figure-ground segregation—all essential for precise grasping when the target is surrounded by flankers. Our results suggest that action planning proactively enhances specific sensory computations during action preparation.

Keywords

Action planning, Grasp, Perception, Surround suppression, Motor, EEG, SSVEP

Introduction

Beyond shaping motor output, action planning critically influences early sensory processing, even before movement begins (van Elk et al., 2010; Gutteling et al., 2015; Gallivan et al., 2019; Gale et al., 2021b; Gale et al., 2021a; Velji-Ibrahim et al., 2022; Chen et al., 2023).

For instance, fMRI studies have shown that information about the intended effector (e.g., limb, eye) and action goal (e.g., grasp, reach, point, or align) can be decoded from activity patterns in early visual cortex (Gutteling et al., 2015; Gallivan et al., 2019; Velji-Ibrahim et al., 2022), as well as in auditory and somatosensory cortex (Gale et al., 2021b; Gale et al., 2021a), during movement planning. Moreover, there is also evidence showing enhanced feature decoding during action preparation (Gutteling et al., 2015; Velji-Ibrahim et al., 2022).

While these studies mainly focused on the modulation of response to a single target object, it remains unclear how action planning modulates spatial interactions between the target and flankers in a cluttered scene. The action-dependent representation of visual features (Lee et al., 2024, 2025), changes of attention allocation (Baldauf and Deubel, 2008, 2010), and the shifted receptive field for saccadic eye movements reported in animal (Neupane et al., 2016) and human (Ruff et al., 2006) studies all suggest that the motor system may proactively tune sensory systems to enhance sensory processing for the imminent actions. In this study, we directly tested whether action planning also tunes the response to a target object in a cluttered scene in a manner demonstrably beneficial for the intended action.

Specifically, to provide this mechanistic test, we focused on visual surround suppression—a fundamental cortical mechanism where the neural response to a central target is typically attenuated by surrounding stimuli (Allman et al., 1985; Cavanaugh et al., 2002). This process is the engine behind edge detection, as it effectively silences neural activity in uniform areas and "highlights" where things change (e.g., change between center and surround; **Fig. 1A**). Therefore, it is pivotal for enhancing visual contrast, sharpening edge detection (Grigorescu et al., 2003, 2004; Nishimura and Sakai, 2004; Lin et al., 2022), and therefore, facilitating figure-ground segregation and object recognition (Palmer, 1999). Such segregation is also indispensable for accurately interacting with objects, particularly in cluttered visual environments like those often encountered during grasping (Binsted et al., 2007; Bulakowski et al., 2009; Chen et al., 2015b; Chen et al., 2015a), making surround suppression an ideal candidate for testing the effect of action planning on spatial interaction in a cluttered scene.

Given that accurately grasping the central target amidst flankers necessitates precise boundary segregation, we predicted that planning a grasp would evoke stronger surround suppression compared to planning a target-irrelevant key release task, where such segregation is not required. Furthermore, dominant and non-dominant hand actions rely on distinct neural control mechanisms, with the former relying more on predictive feedforward control and the latter on impedance-based feedback control (Sainburg, 2002; Gonzalez et al., 2008; Flindall et al., 2013; Tang et al., 2014; Chen et al., 2015b). Consistent with this asymmetry, our previous evidence suggests that surround suppression when *executing* a grasping movement is greater for the dominant than the nondominant hand (Chen et al., 2014; Chen et al., 2015b). If action planning proactively tunes sensory systems, we predicted that this modulation would scale with these distinct control demands, specifically showing enhanced suppression for the more automatized dominant hand.

To test these predictions, we used frequency-tagged electroencephalography (EEG) to track neural responses to the flickering target among flankers via steady-state visual evoked potentials (SSVEPs (Regan, 1977; Vialatte et al., 2010)) during action planning. Right-handed participants, assigned to either dominant or non-dominant hand groups,

planned the target-relevant grasping task or the target-irrelevant key-release task. In addition, to further investigate the action-specificity of the modulation of grasp planning on surround suppression, we also examined the surround suppression during planning a perceptual size-estimation task.

Methods

Participants

Thirty-six participants took part in the study. Seventeen of them (11 females and 6 males, aged 18 -24 with a mean age of 21.45 years old) performed tasks with their dominant hand, and the other nineteen participants (11 females and 8 males, aged 18-29 with a mean age of 21.43 years old) performed the same tasks with their nondominant hand. Our sample size (N = 36) exceeds the minimum of 34 estimated via G*Power for a mixed-design ANOVA with 2 groups and 2 measurements (effect size = 0.25, power = 0.8) (Faul et al., 2007; Faul et al., 2009). We employed a between-subject design specifically to prevent interlimb transfer of learning (Sainburg and Wang, 2002; Stöckel et al., 2016; Rezaei et al., 2025), which could obscure potential differences between the dominant and nondominant hands. This design choice was validated by a control experiment (N = 8, 3 males and 5 females, aged 23-27, with a mean age of 24.75 years), in which they performed tasks with both the nondominant and the dominant hands in a random order, i.e., a within-subjects design. The control experiment replicated the main effect of Task but found no main effect of Hand (dominant vs. nondominant hand), likely due to transfer interference. Moreover, the between-subjects design also served to reduce fatigue effects from the long testing duration. All participants had normal or corrected-to-normal vision, with no history of psychiatric or neurological disorders. All are right-handed as confirmed with the Edinburgh handedness questionnaire (Oldfield, 1971). They all gave written informed consent. The study was approved by the research ethics committee of South China Normal University.

Stimuli

The stimulus, apparatus, and design were exactly the same for the two groups of participants. There were two stimulus conditions to examine the effect of the task on visual

processing. In the Single condition, a target of 4 cm or 5 cm in diameter was presented in isolation in the center of a white background. The size of the target was random on each trial so that participants could perform task according to the size of the target. But in the analysis the stimulus *size* manipulation was collapsed, as it was included only to maintain engagement, not as a factor of interest. The target was flickering on-off at 15 Hz. In the Surrounded condition, the same target was surrounded by 5 flankers of 4 cm in diameter located at the vertices of an equilateral pentagon. The center-to-center distance between the flanker and the target was 6 cm. Because of the similar target and flanker sizes, standard perceptual size-contrast effects (i.e., Ebbinghaus illusion) were minimal or absent (Massaro and Anderson, 1971; Roberts et al., 2005). The target and flankers were flickering on-off at 15 Hz and 12 Hz independently (**Fig. 1B**). The difference in the signals evoked by the target in the Single and Surrounded conditions would reflect the influence of the flankers on the central target, which would also vary with the different tasks.

Importantly, a transparent real disk (thickness: 1 cm) of the same size as the target was placed on top of the flickering target to enable real actions on the physical target for the grasping task (**Fig. 1C**). This is vital, as grasping real objects versus images engages distinct neural mechanisms, particularly within the anterior intraparietal sulcus (aIPS), an area specialized for grasping (Culham et al., 2003; Freud et al., 2018). There was a fixation cross on the center of the screen/target. Participants needed to maintain fixation on the cross throughout the experiment.

Apparatus

The visual stimulus was presented on a computer monitor (Lenovo, size 27 cm × 48 cm, resolution, 1920×1080 pixels, refresh rate 60 Hz). The monitor was placed flat on the desk. Participants' head was stabilized by a chinrest. The physical target (i.e., the transparent disk) was placed right on top of the target on the monitor (**Fig. 1C**). The distance between the chinrest and the desk was 60 cm. The presentation of stimuli was controlled by Psychtoolbox 3 (Brainard, 1997) embedded in MATLAB 2014 (MathWorks Inc; <https://www.mathworks.cn/>).

The EEG data were collected using a NeuroScan system ([Compumedics Neuroscan](https://compumedicsneuroscan.com/); <https://compumedicsneuroscan.com/>) with a 64-channel EEG cap based on the international 10-20 system. A vertical electrooculogram (VEOG) was recorded using two electrodes placed above or below the left eye. A horizontal electrooculogram (HEOG) was recorded from two electrodes placed at the outer canthus of the left and right eyes. The electrode impedance was kept below 10 k Ω . EEG was amplified with a gain of 500 K, band pass filtered at 0.05 – 100 Hz, and digitized at a sampling rate of 1000 Hz. All channels were referenced online to the tip of the nose.

Design

To examine the effect of the grasp planning on the visual processing of the target, a group of right-handed participants were instructed to plan to release the start key and grasp the central physical target (Key release + Grasp, KR+G) with their right-hand before the “go” cue and then execute the task after the “go” cue (**Fig. 1D**). An action task that does not require direct interactions with the target was also included, in which participants just needed to plan to release the start key (Key release, KR) before the “go” cue and then release the start key with their right hand (dominant hand) after the “go” cue (**Fig. 1E**). The contrast between the “Key release + Grasp” and “Key release” would indicate how the planning of action interactions with the target (i.e., target relevance) modulates the inhibition of the flankers on the target.

To examine whether or not the spatial inhibition also depends on the dominance of the hand performing the task, another group of participants performed the KR and KR+G tasks with their left hand (nondominant hand).

To examine whether or not the effect observed above is action-specific, the same two groups of participants also performed a key release and size estimation task (Key release + Estimation, KR+E), in which participants were asked to plan to release the start key and then estimate the perceived size of the target by opening their thumb and index fingers a matching amount before the “go” cue and then execute the task after the “go” cue (**Fig. 1E**). In addition, the KR+E task also helps to rule out the possibility that the difference between

219 KR+G and KR was simply due to the addition of a second task (i.e., grasp) in KR+G that
220 employed more cognitive load than in the KR task.

221 *Procedure*

222 Each trial started with a dashed circle of 4 cm or 5 cm on the screen to remind the
223 experimenter of the size of the target. After the experimenter placed a transparent 3D disk
224 of the same size as the target circle on the target location, the experimenter pressed the
225 “Enter” key to trigger the trial. The participants were instructed to hold down the start key
226 (i.e., the “J” key) at the beginning of each trial.

227 After the participants held down the start key, a red fixation cross appeared in the center of
228 the screen, together with a task cue (“KR”, “KR+G” or “KR+E”), indicating the type of task
229 that the participants had to perform after receiving the “go” cue (see below). Then the
230 stimuli were presented and flickered for 5 sec. Ever since the stimuli started to flicker,
231 participants needed to prepare to perform the action (i.e., “Plan”) but not physically move
232 their hands. The Plan phase could last 1 sec, 2.5 sec, or 4 sec, randomly selected for each
233 trial. After that, the red fixation was changed to green (i.e., “go” cue). Participants needed to
234 perform the task as soon as the fixation color changed (“Execution”). If participants failed to
235 hold down the start key or release the start key before the “go” cue, the trial would stop and
236 be tested again.

237 Each participant performed a practice session before the formal experiment to ensure that
238 they understood the experimental requirements. The order of the 6 conditions (i.e., 2
239 Stimulus conditions × 3 Tasks) were randomized for each participant who either performed
240 the task with their dominant or nondominant hand. Each participant performed 10 runs, 36
241 trials per run. In each run, there were 6 trials for each of the 1-sec and 2.5-sec intervals with
242 the condition randomly selected from the 6 conditions, and 24 trials for the 4-sec interval
243 with 4 trials for each of the 6 conditions. To ensure stable frequency results, only trials with
244 4-sec planning phase were included for analysis and we collapsed trials across the two
245 stimulus sizes, which were manipulated solely to maintain engagement. Therefore, there
246 were 4 trials for each of the 6 conditions in each of the 10 runs, which yielded 40 trials per

participant for each of the 6 conditions of interest. It took each participant about 2 hours to finish the study.

In the control experiment, the stimuli, procedure, and protocol were identical to those of the main experiment, except for the within-subject design (i.e., the same participants performed the tasks with both dominant and nondominant hands) and for the inclusion of only two tasks (KR and KR+G). The same analyses as the main experiment were performed.

Data analysis

EEG data Preprocessing

The EEG data preprocessing was performed with the Letswave Toolbox (<https://letswave.cn/>) embedded in MATLAB (The MathWorks, MA, <https://www.mathworks.com>). First, the data quality was visually inspected. Due to damage, the FP1 data were interpolated from its 3 nearest neighbors: FPz, AF3, and AF7. After channel rejection and interpolation, whole-trial EEG data were re-referenced to the average of all electrodes. The continuous EEG data were filtered offline using a band-pass filter from 1 to 40 Hz. Next, only the trials with a 4-sec action-planning duration were included in the analysis. The segments from 500 ms to 4000 ms after the stimulus onset were extracted. The segments from stimulus onset to 500 ms after stimulus onset were discarded to avoid onset transients (Vergeer et al., 2018). An experimenter sat by the participant to place the object, control the onset of the trial, and monitor the task performance. Trials in which the participants failed to perform the task properly (i.e., failed to shape their hand or performed the wrong action) were rejected. The mean rejection rate for the dominant-hand group was $3.30\% \pm 3.16\%$ (mean $\pm$ SD), and for the nondominant-hand group was $4.60\% \pm 3.59\%$. The remaining data were averaged separately for each condition.

Amplitude of the SSVEPs

The averaged data were then submitted for fast Fourier transformation (FFT), which yielded the steady-state visual evoked potentials (SSVEPs) for each participant and condition at each frequency. Consistent with previous studies (Landau et al., 2010; Vergeer et al.,

2018), electrodes at the parieto-occipital scalp had the largest SSVEPs. Therefore, we averaged the SSVEPs from the following electrodes: PO3, PO4, POZ, O1, O2, and OZ. All the subsequent analyses were based on the averaged SSVEPs from these electrodes. The amplitudes at 15 Hz for the target and 12 Hz for the flanker were extracted for each participant under each condition and were used for the following analyses. Because the electrodes with the largest SSVEPs appeared right-lateralized in the nondominant-hand group, we also performed the same analyses on the 5 electrodes (OZ, O1, POZ, PO4, PO8) that were slightly right-lateralized and showed the largest SSVEPs in this group.

Surround suppression index (SSI)

To quantitatively evaluate the influences of the flankers on the response to the target during the planning of different tasks, we calculated a surround suppression index (i.e., SSI) following previous studies (Chen et al., 2014; Self et al., 2014; Chen et al., 2016; Li et al., 2024). SSI was defined as the ratio of the difference between the SSVEP amplitude of the target under the Single condition and under the Surrounded condition to the SSVEP under the Single condition.

$$SSI = \frac{(SSVEP_{Single} - SSVEP_{Surrounded})}{SSVEP_{Single}}$$

Surround suppression depends on the size and contrast of objects, the distance between target and flankers, the receptive field size of neurons, and the attentional resolution (Xing and Heeger, 2001; Cavanaugh et al., 2002; Li et al., 2024). A positive SSI indicates that the response to the target was inhibited by the flankers in the Surrounded condition compared to the Single condition (i.e., surround suppression). A negative SSI indicates surround enhancement, which was less common than surround suppression but was also reported previously (Xing and Heeger, 2001; Yu et al., 2001).

Time course of SSVEPs

To examine the time course of the effect of grasp planning, we calculated the time course of the SSVEP amplitude at the target flickering frequency (i.e., 15 Hz) for each condition. The epochs included the signals from 0.5 sec before the stimulus onset to 4 sec after the

stimulus onset and then averaged across trials. They were then bandpass filtered with a cutoff transition (low cutoff frequency = 14 Hz; high cutoff frequency = 16 Hz). Then, the amplitude of the target frequency was extracted as the squared magnitude of the result of the Hilbert transform for each condition (Zhang et al., 2018; Davidson et al., 2020).

Statistical analysis

To test whether or not grasp planning modulates surround suppression in a hand-dependent manner (*Primary Hypothesis*), a two-way mixed-design ANOVA with Task (KR, KR+G) as a within-subjects factor and Hand (Dominant, Nondominant) as a between-subjects factor was performed to examine the effect of task relevance and hand dominance on SSIs. To test whether there was a significant surround suppression in each condition, two-tailed one-sample t-tests were performed to compare the SSIs with zero.

To test if the modulation on surround suppression is because of the additional task component in KR+G relative to KR or whether it is specific to action vs. perception (*Control Hypothesis*), we first performed a two-way mixed-design ANOVA on the SSI, with Task (KR+G, KR+E) as a within-subjects factor and Hand (Dominant, Nondominant) as a between-subjects factor to compare the effect of the grasping and the perceptual tasks for both hand groups. KR+E also had two components but it is more perceptual. If KR+G induced more surround suppression than KR+E, then it would suggest that the enhanced surround suppression was not just because of the additional task but is specific to action planning. Similar t-tests were performed to compare the SSI in the KR+E condition and zero.

We also performed a third two-way mixed-design ANOVA on the SSI, with Task (KR, KR+E) as a within-subjects factor and Hand (Dominant, Nondominant) as a between-subjects factor, to examine whether or not KR+E also enhances surround suppression compared to KR. If KR+E did not enhance surround suppression, then it provides further evidence for the action-specific enhancement on surround suppression. Post-hoc paired comparisons were performed to reveal any differences between conditions if any interaction was significant (Bonferroni corrected).

We chose to run separate ANOVAs because our study was designed with a specific hierarchical structure of a priori planned comparisons: first the primary hypothesis and then the control hypothesis. The p values of the main effects of ANOVAs were compared to the most conservative Bonferroni correction threshold for three independent ANOVA tests ($0.05/3 = 0.017$).

To demonstrate the time course of the influence of task planning on the SSVEPs to the target, we compared the time courses of SSVEP point-to-point t-test between the Single and Surrounded conditions, and the results were corrected with a cluster-based permutation test (Zhang et al., 2023).

To demonstrate whether or not the effect of task planning modulates the response to a single target, a two-way ANOVA with Task (KR, KR+G, KR+E) and Hand (Dominant, Nondominant) as factors was performed on the SSVEP amplitude of the target in the Single condition. Post hoc paired comparisons were performed to reveal differences between conditions when the interaction was significant (Bonferroni-corrected). In addition, the time courses of SSVEPs were compared between tasks using a point-to-point t-test corrected with a cluster-based permutation test (Zhang et al., 2023). Similar analyses were also performed to examine whether or not Tasks and Hands modulated the *flanker* response (i.e., 12 Hz) in the Surrounded condition. All statistical analyses were done with JASP (Version 0.95.3; <https://jasp-stats.org/>).

Results

Target relevance and hand dominance additively modulate surround suppression

Consistent with our predictions (**Fig. 2A**), both the nature of the planned action (i.e., whether or not the action is target-related) and hand dominance influenced surround suppression. **Figs. 2B** and **2C** illustrate the target SSVEP amplitudes and their scalp topography for nondominant and dominant hand groups, respectively, while **Fig. 2D** presents the surround suppression indices (SSIs) for all conditions.

A two-way mixed-design ANOVA on the SSI, with Task (KR, KR+G) as a within-subject factor and Hand (Dominant, Nondominant) as a between-subject factor (**Fig. 2D**), revealed several significant main effects. First, planning a grasp (KR+G) elicited a larger SSI than planning a key release (KR) ($F(1, 34) = 21.549, p < 0.001, \eta_p^2 = 0.388$). Furthermore, planning actions with the dominant hand resulted in a larger SSI compared to the nondominant hand ($F(1, 34) = 8.373, p = 0.007, \eta_p^2 = 0.198$). Importantly, there was no significant interaction between Task and Hand ($F(1, 34) = 0.641, p = 0.429, \eta_p^2 = 0.019$), indicating that target relevance and hand proficiency additively enhanced surround suppression.

This additive effect is highlighted by the pattern of SSI values (**Fig. 2D**): suppression was most pronounced when planning a dominant-hand grasp (KR+G), which was significantly greater than zero ($t(16) = 4.245, p = 0.001$). Conversely, suppression was weakest when planning a nondominant-hand key release (KR) ($t(18) = -2.151, p = 0.045$, significantly smaller than 0, i.e., showing surround enhancement). The SSIs for planning a non-dominant-hand grasp ($t(18) = 2.018, p = 0.059$) and a dominant-hand key release ($t(16) = 1.644, p = 0.120$) fell in between these extremes, showing trends towards significant suppression.

The primary analyses reported above utilized a symmetrical electrode montage (PO3, PO4, POZ, O1, O2, and OZ). However, given that the topographical distribution for the nondominant (left) hand appeared right-lateralized (**Fig. 2B**), we sought to verify that our results were not dependent on electrode selection. We re-calculated the surround suppression index using the five electrodes exhibiting the highest peak amplitudes for the nondominant hand, which were slightly right-lateralized (OZ, O2, POZ, PO4, and PO8).

Crucially, the pattern of results remained consistent with our primary findings. First, grasp planning (KR+G) continued to elicit significant surround suppression ($t(18) = 2.595, p = 0.018$). Second, and most importantly, the SSI remained significantly higher in the KR+G task than in the KR task ($t(18) = 3.474, p = 0.003$). Notably, in this lateralized analysis, the KR task did not show significant suppression or enhancement ($t(18) = 1.556, p = 0.137$).

This suggests that the negative SSI (surround enhancement) observed in the symmetrical

montage is likely sensitive to topographical variations, whereas the modulation of suppression by grasp planning is a robust phenomenon.

In the control experiment (within-subject design), there was a significant main effect of Task ($F(1,14) = 9.03$, $p = 0.009$), with the Grasping task (KR+G) eliciting significantly stronger surround suppression than the Key Release (KR) task. A one-sample independent t-test showed that the surround suppression index was significant only in the KR+G task with the dominant hand ($t(7) = 3.1$, $p = 0.017$). These findings confirm that our core finding—that grasp planning modulates visual processing—is robust and reproducible within the same individuals. The main effect of Hand was not significant ($F(1,14) = 0.664$, $p = 0.429$). We believe this result is attributable to interlimb transfer—a well-documented phenomenon where motor skills and neural states associated with one hand transfer to the other (Sainburg and Wang, 2002; Stöckel et al., 2016; Rezaei et al., 2025). In a within-subject design, the neural "set" for the dominant hand likely influences the non-dominant blocks, washing out the dominance effect.

Surround suppression is not modulated by planning a perceptual report task

To ensure that the observed enhancement of surround suppression during grasp planning (KR+G) was genuinely related to the action's nature and not merely due to increased task complexity compared to the simpler key release (KR) task, we introduced a critical control: a perceptual size estimation task (KR+E; **Fig. 1D**). In this task, participants planned to release the start key and then indicate the perceived size of the target by opening their thumb and index finger a matching amount beside the stimuli on the table. Although this involved a similar finger movement to grasping, the KR+E task is more perceptual and relies on more ventral stream: task success depends on the accuracy of the final finger aperture representing perceived size, not on motor programming for object interaction. This type of manual estimation has been instrumental in dissociating the visual ventral stream (perception) from the dorsal stream (action) (Goodale et al., 1991; Chen et al., 2015a; Chen et al., 2018; Zheng et al., 2025).

Consistent with the perceptual nature of the KR+E task, planning this size estimation did not significantly modulate surround suppression. As shown by the SSVEP topographies and amplitudes (**Figs. 3A-3C**), the surround suppression index (SSI) during KR+E planning was not significantly different from zero for either the nondominant hand ($t(18) = 0.624$, $p = 0.541$) or the dominant hand ($t(16) = -1.780$, $p = 0.094$; **Fig. 3D**). This starkly contrasts with the significant suppression observed during grasp planning.

A direct comparison using a two-way mixed-design ANOVA on SSI with Task (KR+G vs. KR+E) and Hand (Dominant vs. Nondominant) revealed a significant main effect of Task ($F(1, 34) = 21.375$, $p < 0.001$, $\eta_p^2 = 0.386$), with stronger SSI for KR+G than for KR+E. Moreover, the two-way mixed-design ANOVA on SSI with Task (KR+E and KR) and Hand (Dominant vs. Nondominant) as factors suggests that the SSI was not significantly different between the KR+E and KR tasks (the main effect of Task was not significant: $F(1, 34) = 0.976$, $p = 0.330$, $\eta_p^2 = 0.028$). All these findings suggest that the enhanced surround suppression observed during dominant-hand grasp planning is not a general consequence of planning a two-component task, but rather a specific neurophysiological adjustment tied to preparing a goal-directed action.

Dominant-hand grasp planning induces sustained surround suppression throughout the preparatory period

To determine whether the planning-related modulation operates transiently or as a sustained state throughout action preparation, we examined the time course of target SSVEP amplitudes (14-16 Hz) in the Single and Surrounded conditions (**Fig. 4A** shows the schematic). For actions planned with the nondominant hand (**Fig. 4B**), significant surround suppression (i.e., lower SSVEP amplitude in Surrounded vs. Single conditions) was observed only sparsely across the planning interval for all three tasks (KR, KR+G, and KR+E; see **Fig. 4B** for specific time windows).

In stark contrast, when planning a dominant-hand grasp (KR+G), surround suppression emerged rapidly and was sustained for nearly the entire planning phase. Specifically, the

target SSVEP amplitude in the Surrounded condition was significantly lower than in the Single condition from approximately 121 ms post-stimulus onset until the end of planning, with only a brief interruption (between 948 ms and 1164 ms; **Fig. 4C**, middle panel). This sustained suppression was unique to dominant-hand grasp planning; for dominant-hand KR and KR+E planning, significant suppression occurred only in scattered, transient intervals (**Fig. 4C**, left and right panels).

These temporal dynamics align with our overall SSI findings but crucially reveal that planning a dominant-hand grasp engages a robust and continuous modulation of surround suppression. This suggests that planning-related modulation for demanding actions is not a fleeting preparatory event but rather a persistent state of heightened visual processing that begins shortly after stimulus presentation and endures throughout the preparation period.

Sparse evidence for action planning modulating neural responses to an isolated target

Our findings thus far demonstrate that action planning modulates surround suppression. However, since surround suppression is calculated as the difference in target response between isolated (Single) and surrounded conditions, it is conceivable that action planning did not actually alter the suppressive influence of the flankers per se, but instead independently modulated the neural response to the isolated target. Previous behavioral work, for instance, has shown that action preparation can facilitate target-related performance even for isolated targets (Gutteling et al., 2011; Mason et al., 2015).

To address this alternative explanation, we directly examined whether planning different actions (KR, KR+G, KR+E) influenced the SSVEP response to the target presented in the Single (isolated) condition. Importantly, we found sparse evidence for such modulation. Specifically, the main effects of Tasks and Hands were not significant (Task: $F(2, 68) = 0.108$, $p = 0.898$, $\eta_p^2 = 0.003$; Hand: $F(1, 34) = 2.275$, $p = 0.141$, $\eta_p^2 = 0.063$). The interaction between them was not significant ($F(2, 68) = 0.323$, $p = 0.725$, $\eta_p^2 = 0.009$). This is also confirmed in the comparison between the time courses of the amplitudes (**Fig.**

5). Specifically, we found that, for the single target, the only significant difference was observed between the KR+G and KR+E conditions for a short period from 1800 ms to 2620 ms after stimulus onset in the dominant-hand group.

These results indicate that the observed influence of grasp planning in our study is independent from a general enhancement or attenuation of the neural response to the target itself. Instead, action planning specifically modulates the *interaction* between the target and its surrounding flankers, enhancing spatial inhibition to the target to facilitate target-flanker segregation, particularly in preparation for a grasp.

Grasp planning does not affect the response to the flankers

We also analyzed the response to the flankers (i.e., 12 Hz) in the Surrounded condition (no flankers were presented in the Single condition; **Fig. 6**). The amplitudes for the flankers were generally smaller than the amplitudes for the target, probably because the flankers were presented in the periphery (Phangwiwat et al., 2024). Two-way mixed design ANOVA with Task (KR, KR+G or KR+E) as a within-subjects factor and Hand as a between-subjects factor showed that none of the main effects or the interaction was significant (Main effect of Task: $F(2, 68) = 0.556$, $p = 0.576$, $\eta_p^2 = 0.016$; Hand: $F(1, 34) = 0.252$, $p = 0.619$, $\eta_p^2 = 0.03$; Interaction between Task and Hand: $F(2, 68) = 2.003$, $p = 0.143$, $\eta_p^2 = 0.056$). There was no significant main effect or interaction (all $F(1, 34) < 2.693$, $p > 0.110$) even for ANOVAs with two Task levels (KR and KR+G; KR+G and KR+E, or KR+E and KR) and two Hand levels, as we did for the target response above. The independent t-test did not reveal any differences between groups either (Dominant vs. Nondominant: KR, $t(34) = 0.004$, $p = 0.997$; KR+G, $t(34) = 0.821$, $p = 0.417$; KR+E, $t(34) = 0.6662$, $p = 0.512$). Comparison between the time course of the amplitude of flankers did not reveal any difference between any pair of tasks. All these results suggest that the action-planning effect is target-specific. The motor system does not simply "dampen the periphery" globally; rather, it specifically suppresses the response to the center target to enhance the segregation of the goal object.

Discussion

Our study provides compelling neural mechanistic evidence that preparing a hand action proactively refines early visual processing, specifically by enhancing visual surround suppression—a key cortical mechanism for object segregation. This modulation was dependent on the action's relevance to the target and the hand used for planning, being strongest for dominant-hand grasps, and notably absent when planning a perceptual report that relies on more ventral stream than the KR+G task. These findings suggest that motor planning may actively tune sensory systems to optimize visual input for efficient goal-directed behavior.

The primate visual system is characterized by extensive feedback projections from higher- to lower-order cortical areas, with connections to early visual cortex like V1, the primary generator of SSVEP signals (Müller et al., 1997; Di Russo et al., 2007; Wittevrongel et al., 2018), potentially outnumbering feedforward inputs (Felleman and Van Essen, 1991; Carandini et al., 2005; Muckli and Petro, 2013). This architecture is known to support diverse top-down functions, including visual working memory, attention, and predictive processing (Rao and Ballard, 1999; Harrison and Tong, 2009; Gilbert and Li, 2013).

Converging evidence further indicates that these feedback pathways also convey motor-related information during action planning (Gutteling et al., 2015; Gallivan et al., 2019; Gale et al., 2021b; Gale et al., 2021a; Velji-Ibrahim et al., 2022). *What* types of action-related information are represented and the precise neural mechanisms underlying this preparatory sensory tuning have remained less clear.

A key contribution of our current work is to illuminate *how* such planning-related modulation might operate at a mechanistic level. By focusing on surround suppression—a well-understood visual cortical mechanism vital for figure-ground segregation—we show that action planning can selectively adjust its gain. By selectively boosting surround suppression, the visual response to the target region was inhibited, which increases the contrast between the target stimuli and similar surround stimuli, making the edges between target and flankers stand out. This targeted modulation serves to improve edge detection

and facilitate figure-ground segregation—computational steps that are critical for precise grasping in cluttered environments (Grigorescu et al., 2003, 2004; Nishimura and Sakai, 2004; Sang et al., 2017; Lin et al., 2022). Thus, we interpret this suppression not as a reduction in signal strength, but as an optimization of the contrast required to define the object for physical interaction.

Although we did not record digital kinematics during the EEG session itself — because introducing motion tracking sensors or data gloves can introduce electrical artifacts during the sensitive planning phase we were analyzing — this limitation does not undermine the interpretation. The planning-related modulation of surround suppression we observe resonate with behavioral studies on visual crowding in the periphery—a phenomenon partly driven by surround suppression (Chen et al., 2014)—which show better size scaling for the surrounded target during grasping than perceptual size estimation. The current neural evidence for enhanced suppression *during planning* suggests this critical perceptual-motor difference is established well before movement onset and likely persists into execution. It should be noted that accurate grip scaling relies on the precise segregation of the object's boundaries rather than the overall response amplitude of the target. Although the neural response to the target was suppressed, our findings suggest that grasp planning inhibits the response to the target area, making the boundary between the target and the flankers more distinct. This enhanced edge information is likely what facilitates the robust grip scaling observed in behavioral crowding studies (Chen et al., 2015b; Chen et al., 2015a), even when the subjective perception of the object's size is compromised.

Our findings provide robust neural evidence supporting the Premotor Theory of Attention (Craigheero and Rizzolatti, 2005), which posits that spatial attention arises from the same neural circuits that program motor actions. Consistent with behavioral evidence that action preparation recruits attentional resources to the goal location (Baldauf and Deubel, 2009), we propose that the observed modulation of surround suppression represents a motor-driven attentional mechanism. Specifically, the intent to grasp triggers top-down signals that sharpen the visual representation of the target object—enhancing figure-ground segregation—thereby optimizing sensory processing for the demands of the upcoming

movement. The effector-specificity of this modulation—stronger surround suppression for dominant-hand planning—and its absence during a comparable perceptual task, strongly point to a direct influence of motor-related signals. The observation that spatial attention directed to the target object within a neuron's receptive field would reduce surround suppression observed in this neuron (Sundberg et al., 2009), an opposite effect of the enhanced surround suppression during the target-directed grasping preparation in our study, also suggests that grasp planning is not a passive reflection of attentional mechanisms *per se* but rather an active, goal-directed process initiated by the motor system. An alternative interpretation is that grasp planning redirects spatial attention to the peripheral flanker region, which could drive center suppression via the peripheral attention mechanism described by Sundberg et al. (2009). Our flanker SSVEP data argue against this account. Grasp planning did not significantly modulate the neural response to the flankers, indicating that the enhanced surround suppression we observe reflects a direct motor-driven modulation of target processing rather than a secondary consequence of attention to the surround.

The modulation of surround suppression also revealed a clear distinction between the grasping and perception systems in visual spatial resolution in the central visual field. While grasping requires precise edge detection and figure-ground segregation, which causes increased surround suppression, the size-estimation perception task also requires edge detection. However, it does not modulate surround suppression, suggesting that there is no requirement for enhancing edge detection possibly because the spatial resolution of the visual system in the central visual field is higher for perception than for grasping during planning. Indeed, previous studies showed that the fovea is overrepresented in the ventral stream for visual perception (Malach et al., 2002), while the entire visual field is more evenly represented the visual-dorsal stream for visually-guided action (Pitzalis et al., 2006).

We did not observe surround suppression during the planning of the size estimation task. While this might initially seem inconsistent with the Ebbinghaus illusion—where perceived size is distorted by flankers—it is important to note that such contextual size-contrast illusions rely heavily on specific size ratios and distances (Massaro and Anderson, 1971;

Roberts et al., 2005). In our study, the flankers and targets were approximately the same size (4 cm vs. 4 or 5 cm; a ~1:1 ratio). Under these specific geometric conditions, standard perceptual size-contrast effects are typically minimal or absent (Massaro and Anderson, 1971; Roberts et al., 2005). Thus, the lack of neural suppression in our perceptual task aligns with the physical stimulus properties.

More fundamentally, the dissociation we observed between the grasping (KR+G) and estimation (KR+E) tasks likely reflects the distinct functional demands of these motor goals. In the grasping task, the flankers act as physical obstacles that must be actively avoided to prevent collision and ensure a stable grip. Consequently, the motor system must actively enhance figure-ground segregation—manifesting as increased surround suppression—to sharply define the target's boundaries relative to these obstacles. In contrast, during the estimation task, the flankers serve merely as visual context; they impose no physical constraint on the manual report. As such, there is no top-down motor requirement to suppress the surround, explaining the absence of modulation in this condition despite the identical visual input. In this view, the null result in the estimation task serves as a critical control that validates the action-specificity of our findings.

In conclusion, by demonstrating that action planning, but not perceptual planning, specifically enhances visual surround suppression, our study provides critical neural mechanistic evidence that motor intentions can translate into concrete changes in early visual processing, optimizing the brain's ability to segregate relevant objects from distractors. This work underscores the profound integration of perception and action, revealing how the motor system actively prepares sensory cortices for the specific challenges of interacting with a complex world.

Data and Software availability

The data and code will be available upon request to the last author at juanchen@m.scnu.edu.cn.

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757

758 **Figure legends**

759 **Figure 1. The stimuli, setup, protocol, and tasks of the study.** (A) Schematic illustration of how
760 surround suppression sharpens edge detection. By suppressing responses in the center target, the
761 contrast between the target and the flankers increases, making the edges between the target and
762 the flankers stand out and facilitating edge detection. (B) The stimuli of the study. In the Single
763 condition, a solid circle was presented at the center of the screen, which was used as the target for
764 the grasping task. In the Surrounded condition, the target circle was surrounded by 5 circles (i.e.,
765 flankers). In both conditions, the target circle flickered at 15 Hz. In the Surrounded condition, the
766 flanker flickered at 12 Hz. A transparent real disk (thickness: 1 cm) of the same size as the target
767 was placed on top of the flickering target to enable real actions on the physical target for the
768 grasping task (see below). (C) The setup of the study. The screen was placed flat on the table. The
769 keyboard was placed next to the screen. A chinrest was used to keep participants' heads stabilized
770 throughout the experiment. (D) The protocol of one trial. Participants held down the start key on the
771 keyboard ("J" key) from the beginning of each trial. There was a dashed circle presented on the
772 screen to remind the experimenter of the size of the target. The experimenter placed a 3D
773 transparent disk of the same size on top of the dashed circle. Then, the task cue appeared to
774 indicate the task of this trial. After that, the stimulus of either the Single or Surrounded condition
775 started to flicker and participants started to prepare for the task ("Plan") while keeping their fingers in
776 the starting position. After the stimulus was flickering for an interval randomly selected from 1 sec,
777 2.5 sec, and 4 sec, the red fixation was changed to green, which served as the "go" cue for the task.
778 On each trial, the flickering of the stimulus lasted for 5 sec in total. Note: only the data in the "task
779 planning" phase in the 4 sec condition was analyzed. The data after the "go" cue appeared were

discarded. **(E)** The three main tasks of the experiment. Participants planned and executed three tasks. In the Key Release (KR) condition, participants just needed to release the start key when received the “go” cue. In the “KR + Grasp” (KR+G) condition, they had to release the start key and grasp the target 3D transparent disk, lift it up and put it down. In the “KR + Estimate” (KR+E) condition, participants were required to release the start key and report the perceived size of the target disk by opening their thumb and index finger a matching amount beside the object on the table. In all conditions, participants needed to maintain fixation on the center cross throughout the experiment.

Figure 2. The predictions and the results of the amplitudes and surround suppression

indices at the planning stage for the KR and KR+G tasks. (A) We predicted that both the target-relevance of the action task (KR, a target-irrelevant task, vs. KR+G, a target-relevant task) and the proficiency of the hand (i.e., hand dominance, nondominant hand vs. dominant hand) performing the task modulate surround suppression. The more target-relevance and hand proficiency, the more surround suppression. **(B)** The results of the nondominant-hand group. Left: the SSVEP amplitude as a function of the frequency around 15 Hz averaged from the six electrodes (PO3, PO4, POZ, O1, O2, and OZ; yellow dots on the map). Right: the SSVEP amplitude of the target frequency (15Hz) for each individual at each condition. **(C)** The results of the dominant-hand group. Similar with **(B)**. **(D)** The surround suppression index (SSI) for the KR and KR+G tasks for both participant groups performing the tasks with either their nondominant or their dominant hands. Each dot represents the result of one participant. * denotes $p < 0.05$. ** denotes $p < 0.01$.

Figure 3. The results of the amplitude and surround suppression indices for the perception

task (KR+E) at the planning stage. (A-C) similar to **(B)** and **(C)** in Figure 2. In **(B)**, Single-D and Surrounded-D indicate results for the visual conditions with the dominant hand, while the Single-NonD and Surrounded-NonD indicate results for the visual conditions with the nondominant hand. **(D)** the SSI for the KR+E task for participants groups performing the tasks with either their nondominant or their dominant hands.

Figure 4. Time course of the amplitude of SSVEPs around the target frequency for each

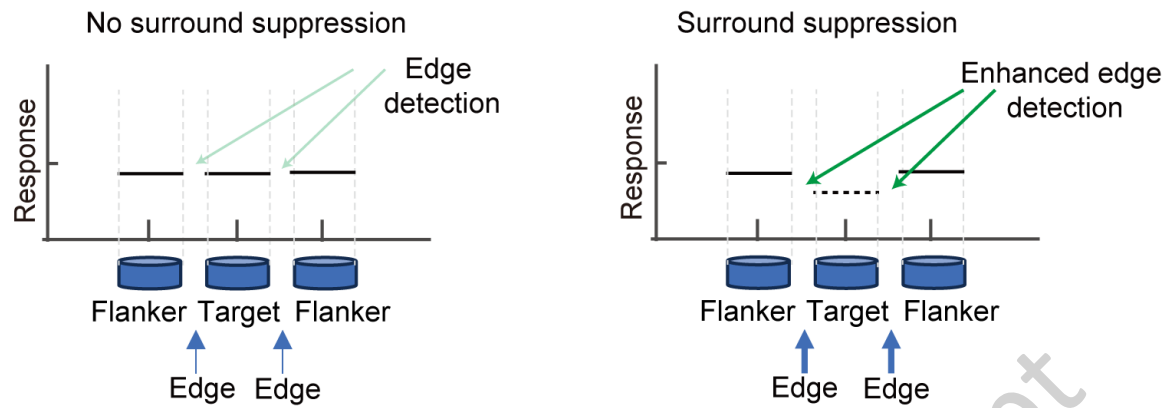
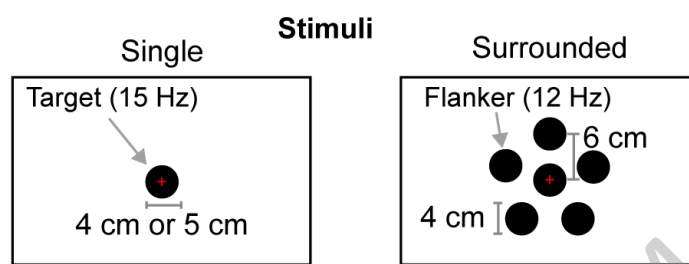
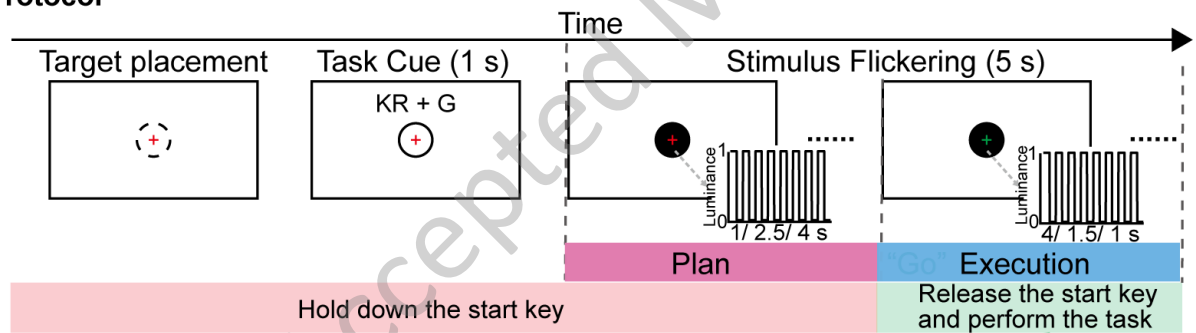
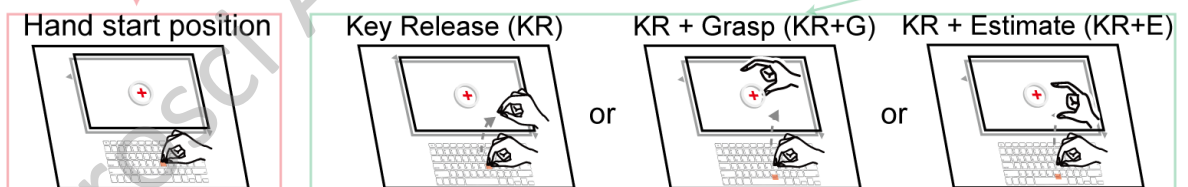
condition. (A) Schematic demonstration of the time course of the amplitude of around 15 Hz (i.e., 14-16 Hz). Red dashed line shows the time course of the amplitude. **(B) and (C)** Shaded areas

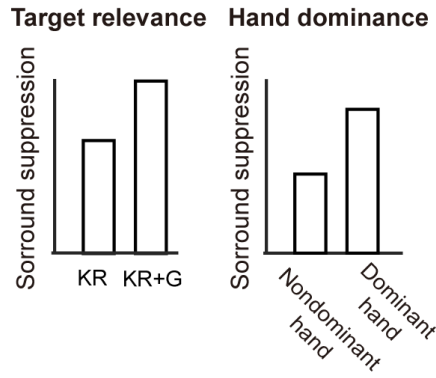
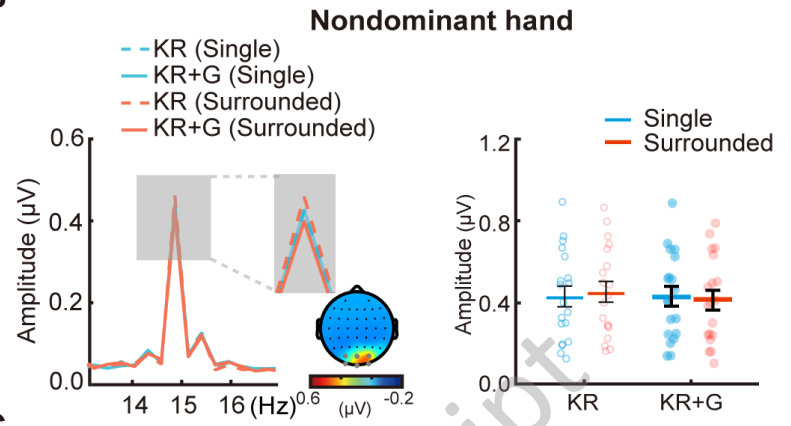
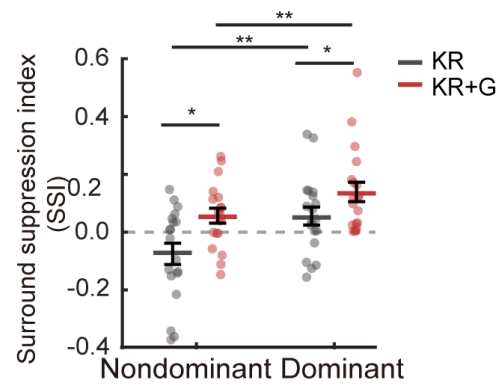
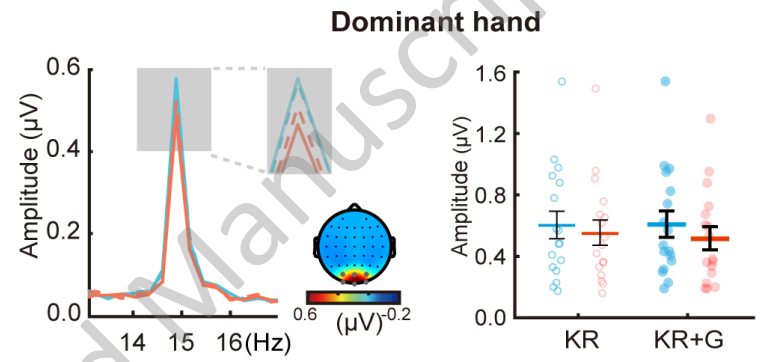
809 show when the amplitude of the target frequency was significantly different between the Single and
810 Surrounded conditions (point to point paired t-test, two-tailed, $p < 0.05$, corrected with cluster-based
811 permutation test) for the nondominant and dominant hands, respectively.

812 **Figure 5. Time course of the amplitude of SSVEPs around the target frequency for each task**
813 **condition when the target was presented in isolation (i.e., Single condition). (A)** results of the
814 nondominant hand group. **(B)** results of the Dominant hand group.

815 **Figure 6. Results of the amplitudes of flankers in the Surrounded conditions for all three**
816 **tasks. (A)** The SSVEP amplitude as a function of the frequency around 12 Hz averaged from the
817 six electrodes (PO3, PO4, POZ, O1, O2, and OZ; yellow dots on the map). **(B)** The SSVEP
818 amplitude of the flanker frequency (12Hz) for each individual at each condition. **(C)** Time course of
819 the amplitude of SSVEPs around the flanker frequency for each condition.

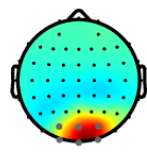
820

A**B****C****D Protocol****E Tasks**

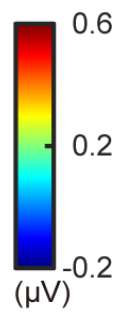
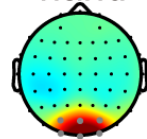
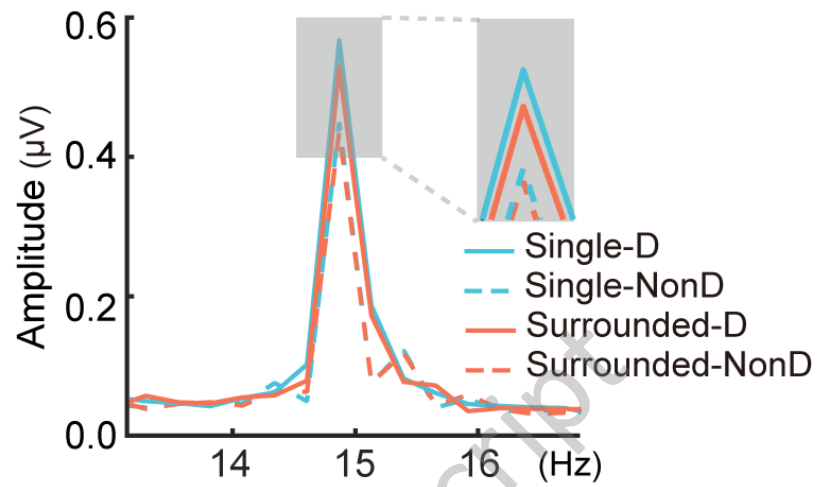
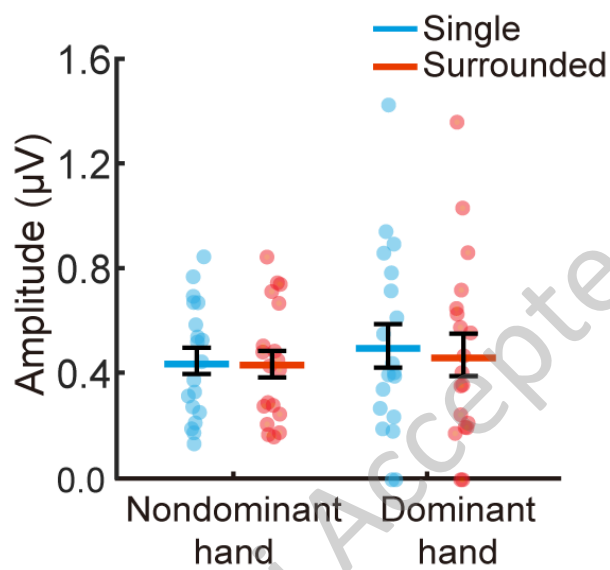
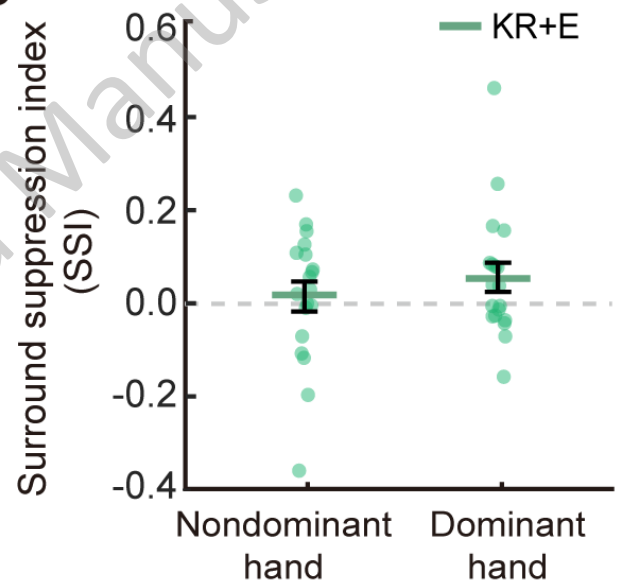
A**B****D****C**

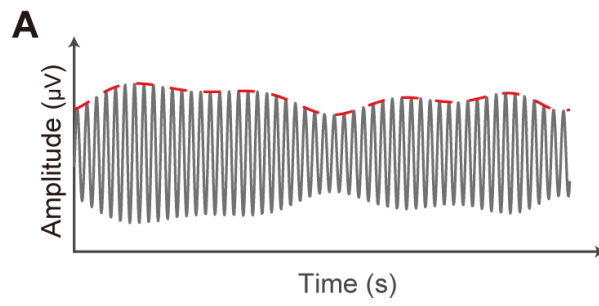
A

Nondominant
hand

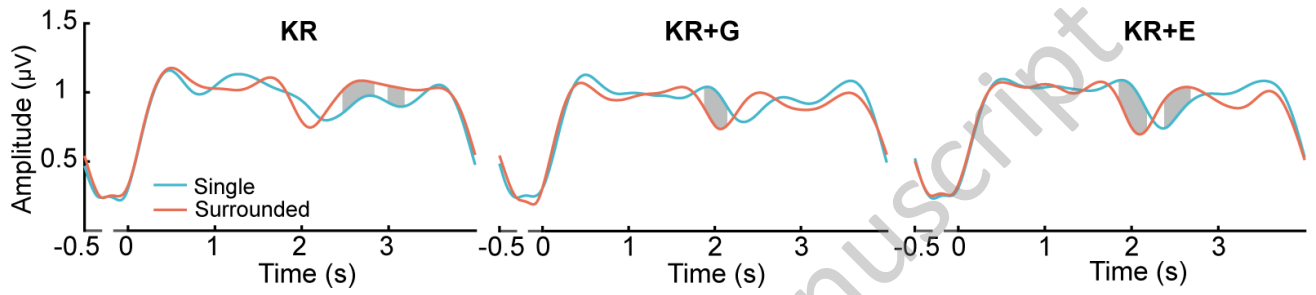


Dominant
hand

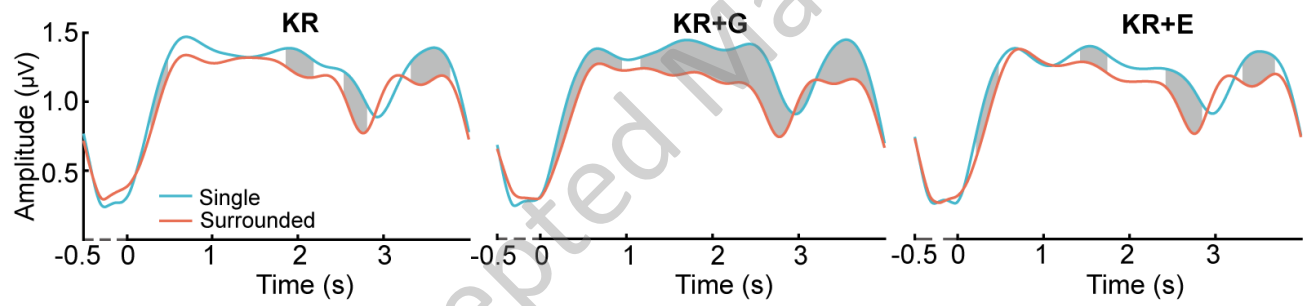
**B****C****D**

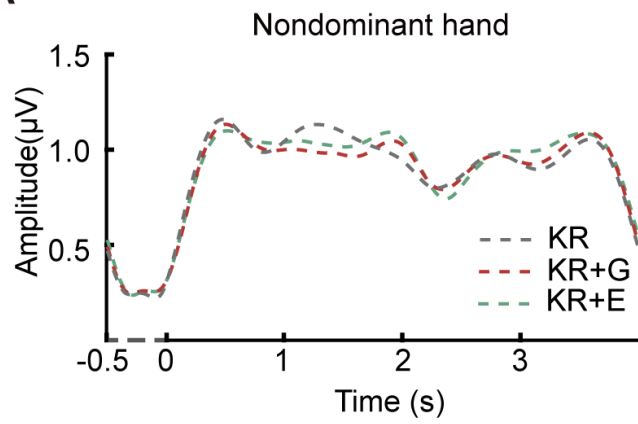
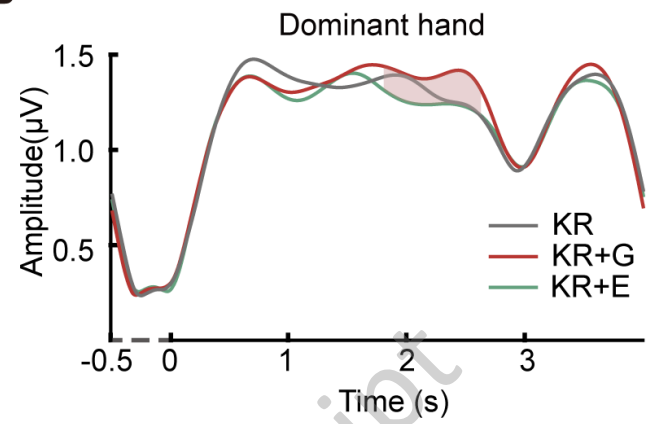


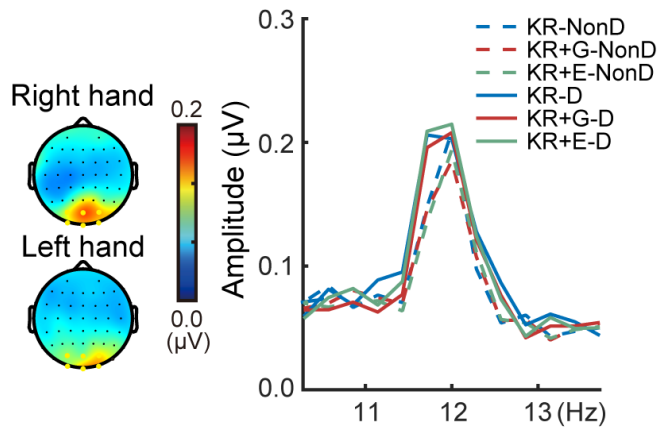
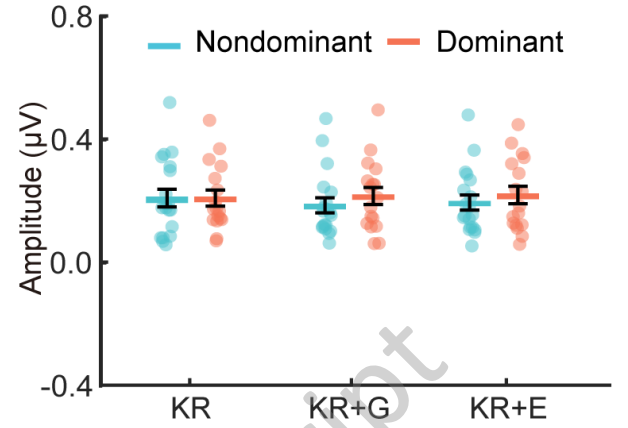
B Nondominant hand



C Dominant hand



A**B**

A**B****C**